

State of Developer Ecosystem 2024

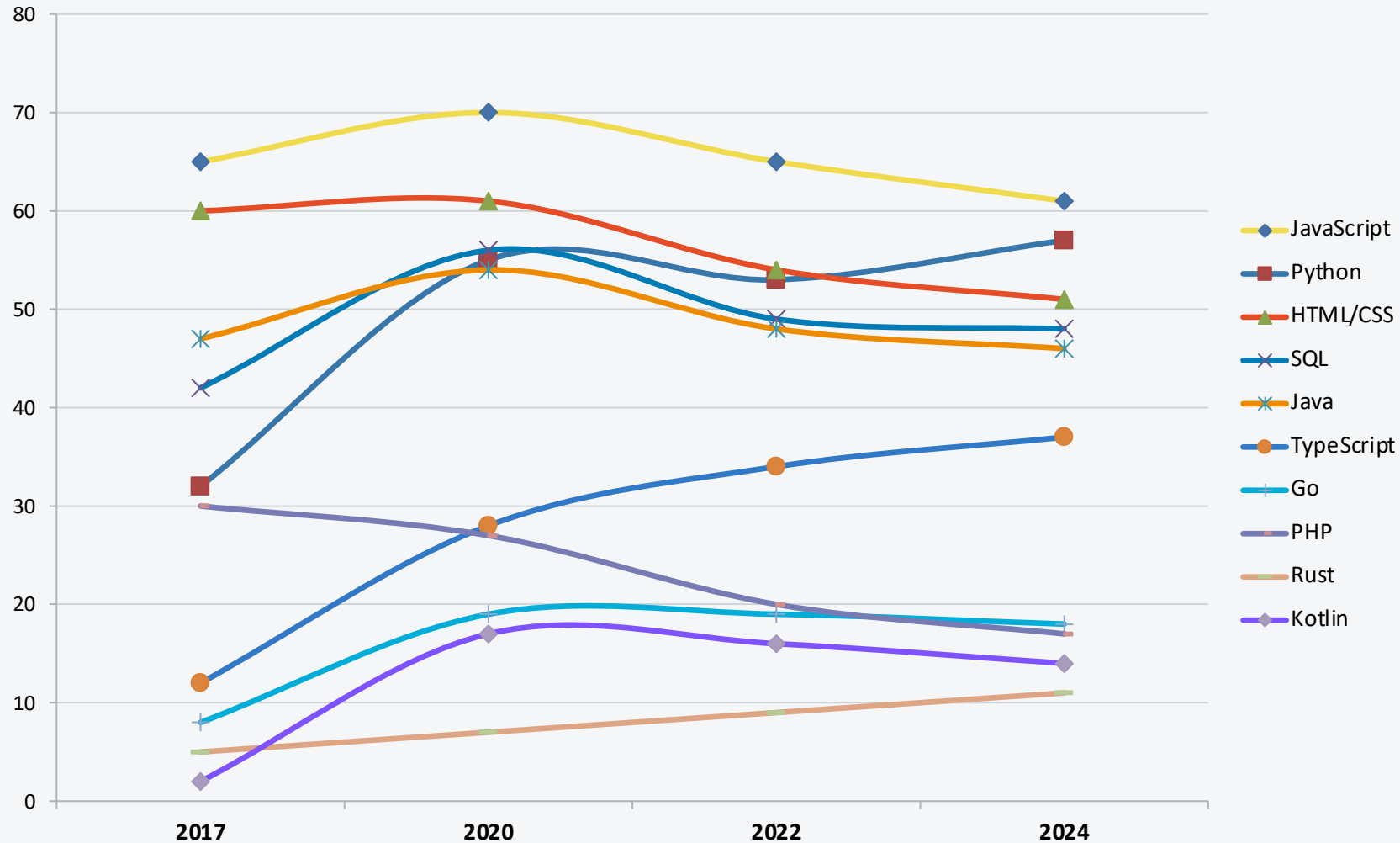
Language Trends, Satisfaction, and Strategic Shifts

Based on JetBrains' annual survey of 23,262 developers worldwide

Covering programming languages, tools, AI attitudes, and career trends across the global developer community

Python Closes the Gap: 8 Years of Language Usage Shifts

Percentage of developers using each language | Source: JetBrains Developer Ecosystem Survey



Key Shifts

JavaScript peaked at 70% in 2020, now at 61%

Python rose from 32% to 57% — nearly closing the gap

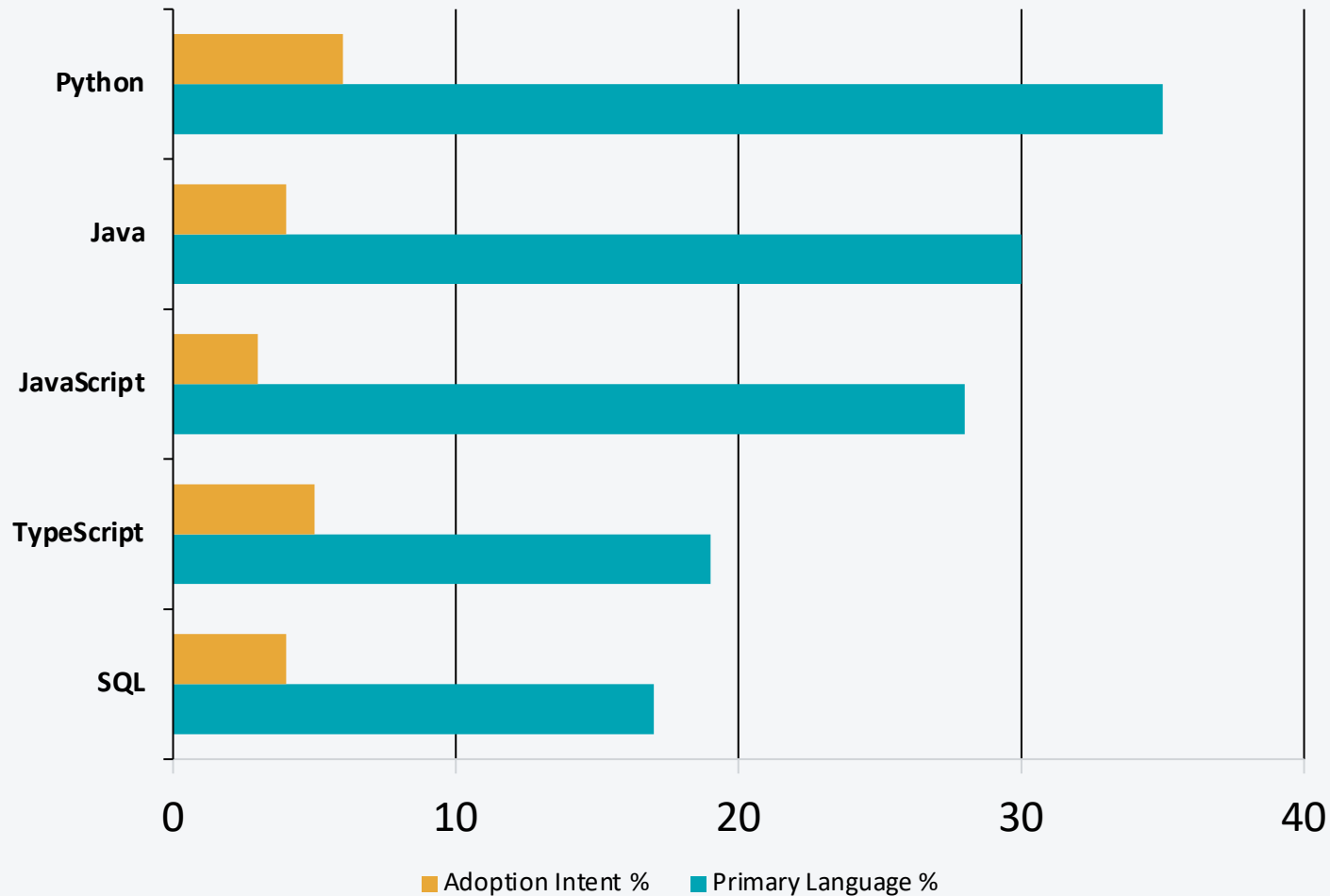
TypeScript tripled from 12% to 37%

PHP dropped from 30% to 17%

Rust growing steadily, now at 11%

Python Is Now the #1 Primary Language at 35%

Primary language usage vs. adoption intent reveals strong ecosystem lock-in



The Comfort Zone Effect

Adoption intent stays below 6% across all languages — developers rarely plan to switch their primary stack.

Python leads as primary language at 35%, even though JavaScript remains most-used overall at 61%.

This gap shows ecosystem lock-in: expertise investment keeps developers loyal to their chosen tools.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Satisfaction scores from JetBrains Developer Ecosystem Survey 2024



What Drives Satisfaction?

TypeScript (199) and Rust (198) dominate satisfaction — both offer type safety and modern tooling.

Python (151) leads among high-adoption languages, showing you don't sacrifice satisfaction for popularity.

Traditional systems languages (C, C++) score lower despite critical use cases.

Pattern: type-safe, modern-tooling languages consistently outrank legacy alternatives.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice is driven by domain, not general preference

Application Domain	Leading Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
Integrating with APIs	Python	38%
Data Science / Analytics	Python	28%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic	Java	55%
System Tools & Components	C++	27%

Domain Dominance

Python owns AI/ML (31%) and data science (28%) — the AI boom directly fuels its growth.

TypeScript/JavaScript dominate UI at 62%/58% — frontend is a two-language ecosystem.

Java leads APIs & services (49%) and application logic (55%) — enterprise backend runs on Java.

C++ leads system tools (27%) — performance-critical infrastructure stays native.

Implication: Hire by domain, not by language preference alone.

What This Means for Your Technology Strategy in 2025

1

Invest in Python for AI/ML Teams

Python is the undisputed leader in AI/ML (31%) and data science (28%). Every AI initiative should be Python-first.

2

TypeScript Is the Future of Frontend & Full-Stack

TypeScript triples to 37% usage and tops satisfaction at 199. Modern web development converges on TypeScript.

3

Rust & Go Are the Growth Plays for Infrastructure

Go (18%) and Rust (11%) address modern infrastructure needs — microservices, cloud-native, and systems programming.

4

PHP & Ruby Hiring Pools Are Shrinking

PHP dropped from 30% to 17%; Ruby from 10% to 4%. Plan succession for legacy codebases in these ecosystems.

5

Satisfaction Correlates with Type Safety & Modern Tooling

Factor developer satisfaction into retention strategy — TypeScript and Rust prove that better tooling drives loyalty.